Week 2 - Programming Assignment-1: Difference

In this assignment, you will have to take two numbers as input and print the difference.

Input Format:

The first line of the input contains two numbers separated by a space.

Output Format:

Print the difference in single line

Example:

Input: 4 2

Output: 2

Explanation: Since the difference of numbers 4 and 2 is 2, hence the output is 2.

Sample Test Cases Input Output Test Case 1 6 2 4 Test Case 2 10 10 0 Test Case 3 12 5 7 Test Case 4 3 5 -2 Test Case 5 100 100 0 Test Case 6 5 10 -5 Test Case 7 5 12 -7 Test Case 8 8 4 4

In [6]:
```
x, y = map(int, input().split())
print(x-y)
```

```
4 2
2
```

or

In [7]:
```
x,y = input().split(" ")
x = int(x)
y = int(y)

print(x-y)
```

```
4 2
2
```

Week 2 - Programming Assignment-2: Large

Given two numbers as input, print the larger number.

Input Format: The first line of input contains two numbers separated by a space

Output Format: Print the larger number

Example:

Input: 2 3

Output: 3

Sample Test Cases Input Output Test Case 1 2 3 3 Test Case 2 3 4 4 Test Case 3 4 5 5 Test Case 4 10 8 10 Test Case 5 100 10 100 Test Case 6 0 1 1 Test Case 7 10 9 10 Test Case 8 9 6 9

In [8]:
```
x, y = map(int, input().split())
print(x if x>y else y)
```

```
2 3
3
```

or

In [9]:
```
x,y = input().split(" ")
x = int(x)
y = int(y)

if(x>y):
    print(x)
else:
    print(y)
```

```
2 3
3
```

Week 2 - Programming Assignment-3: Loops ,List and Sum

You all have seen how to write loops in python. Now is the time to implement what you have learned.

Given an array A of N numbers, you have to write a program which prints the sum of the elements of array A with the corresponding elements of the reverse of array A. If array A has elements [1,2,3], then reverse of the array A will be [3,2,1] and the resultant array should be [4,4,4].

Input Format:

The first line of the input contains a number N representing the number of elements in array A. The second line of the input contains N numbers separated by a space. (after the last elements, there is no space)

Output Format:

Print the resultant array elements separated by a space. (no space after the last element)

Example:

Input: 4 2 5 3 1

Output: 3 8 8 3

Explanation: Here array A is [2,5,3,1] os reverse of this array is [1,3,5,2] and hence the resultant array is [3,8,8,3]

Sample Test Cases

```
            Input | Output
```

Test Case 1| 4 1 2 3 4 5 |5 5 5 Test Case 2| 5 1 2 3 4 5 |6 6 6 6 6

In [11]:
```
N = int(input())
a = [int(x) for x in input().split()]
revA = a[::-1]
result = [None] * len(a)
for i in range(len(a)):
    result[i] = a[i] + revA[i]

for i in range(len(a)):
    print(result[i], end=" ")
```

```
4
2 5 3 1
3 8 8 3
```

or

In [12]:
```
N = int(input())
A = [int(i) for i in input().split(" ")]
B = []
for i in range(len(A)-1, -1,-1):
    B.append(A[i])
C = []
for i in range(len(B)):
    C.append(A[i]+B[i])
for i in range(len(C)):
    if(i==len(C)-1):
        print(C[i])
    else:
        print(C[i],end=" ")
```

```
4
2 5 3 1
3 8 8 3
```

Week 3 - Programming Assignment-1: Max and Min

Given a list of numbers, find maximum and minimum in this list.

Input Format:

The first line contains numbers separated by a space.

Output Format:

Print maximum and minimum separated by a space

Example:

Input:

1 2 3 4 5

Output:

5 1

Sample Test Cases | Input | Output Test Case 1 |2 3 4 5 6|6 2 Test Case 2 |1 3 5 7|7 1 Test Case 3 |10 11 4 1 6 7 2 9|11 1 Test Case 4 |10 11 100 200 300 34|300 10 Test Case 5 |10|10 10 Test Case 6 |1 1 | 1 1 Test Case 7 |1 2 3 4 5 6 7 8 |8 1 Test Case 8 |1 2|2 1

In [13]:
```
a = [int(x) for x in input().split()]
print(max(a), min(a))
```

```
1 2 3 4 5
5 1
```

Week 3 - Programming Assignment-2: Multiple of 3

Given a list A of numbers, you have to print those numbers which are not multiples of 3.

Input Format:

The first line contains the numbers of list A separated by a space.

Output Format:

Print the numbers in a single line separated by a space which are not multiples of 3.

Example:

Input:

1 2 3 4 5 6 5

Output:

1 2 4 5 5

Explanation: Here the elements of A are 1,2,3,4,5,6,5 and since 3,6 are the multiples of 3 hence after removing them the list becomes 1,2,4,5,5. Sample Test Cases | Input | Output Test Case 1 |2 3 4 5 6|2 4 5 Test Case 2 |21 47 32 43|47 32 43 Test Case 3 |8 9 6 5 3|8 5 Test Case 4 |87 34 12 90 34 23 98 67 12 23|34 34 23 98 67 23 Test Case 5 |23 87 56 34 12 61 58 64 29 98 2 16 2 6 9|23 56 34 61 58 64 29 98 2 16 2 Test Case 6 |1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 16 17 18 19|1 2 4 5 7 8 10 11 13 14 16 17 19

In [14]:
```
a = [int(x) for x in input().split()]
for i in range(len(a)):
    if ((a[i]%3)!=0):
        print(a[i], end=" ")
```

```
1 2 3 4 5 6 5
1 2 4 5 5
```

or

In [15]:
```
a = [int(x) for x in input().split()]

b = []

for i in a:
    if(i%3!=0):
        b.append(i)

for i in range(len(b)):
    if(i==len(b)-1):
        print(b[i],end="")
    else:
        print(b[i],end=" ")
```

```
1 2 3 4 5 6 5
1 2 4 5 5
```

Week 3 - Programming Assignment-3: Digits

You are given a number A which contains only digits 0's and 1's. Your task is to make all digits same by just flipping one digit (i.e. 0 to 1 or 1 to 0) only. If it is possible to make all the digits same by just flipping one digit then print 'YES' else print 'NO'.

Input Format:

The first line contains a number made up of 0's and 1's.

Output Format:

Print 'YES' or 'NO' accordingly without quotes.

Example:

Input:

101

Output: YES

Explanation: If you flip the middle digit from 0 to 1 then all the digits will become same. Hence output is YES. Sample Test Cases|Input|Output Test Case 1 |10|YES Test Case 2 |1011|YES Test Case 3 |0011|NO Test Case 4 |11|NO Test Case 5 |10|YES Test Case 6 |0|YES

In [16]:
```
A = input()
ls = []
li = str(A)
for j in li:
    ls.append(int(j))
count_z = 0
count_o = 0
for k in ls:
    if(k==1):
        count_o += 1
    if(k==0):
        count_z += 1

if((count_o == 1) or (count_z == 1)):
    print("YES")

else:
    if((count_o == 0) or (count_z == 0)):
        print("NO")
    else:
        print("NO")
```

```
101
YES
```

Week 4 - Programming Assignment-1: Factorial

Given a number n, you have to print the factorial of this number. To know about factorial please click on this link.

Input Format:

A number n.

Output Format:

Print the factorial of n.

Example:

Input: 4

Output: 24

Sample Test Cases|Input|Output Test Case 1 |5|120 Test Case 2 |4|24 Test Case 3 |6|720 Test Case 4 |8|40320 Test Case 5 |7|5040 Test Case 6 |10|3628800

In [17]:
```
n = int(input())
fact = 1

for i in range(1,n+1):
    fact = fact * i

print (fact)
```

```
4
24
```

or

In [18]:
```
k = int(input())

fac = 1
for i in range(1,k+1):
    if(k==0):
        break
    fac=fac*i

print(fac)
```

```
4
24
```

Week 4 - Programming Assignment-2: Matrix

You are provided with the number of rows (R) and columns (C). Your task is to generate the matrix having R rows and C columns such that all the numbers are in increasing order starting from 1 in row wise manner.

Input Format: The first line contain two numbers R and C separated by a space.

Output Format: Print the elements of the matrix with each row in a new line and elements of each row are separated by a space.

NOTE: There should not be any space after the last element of each row and no new line after the last row.

Example:

Input: 3 3

Output: 1 2 3 4 5 6 7 8 9

Explanation: Starting from the first row, the numbers are present in the increasing order. Since it's a 3X3 matrix, the numbers are from 1 to 9. Sample Test Cases| Input| Output Test Case 1 |2 3|1 2 3 |4 5 6 Test Case 2 |3 2|1 2 |3 4 |5 6

In [19]:
```
a = [int(x) for x in input().split()]
n=1
for i in range(a[0]):
    for j in range(a[1]):
        if(n<=(a[0]*a[1])):
            if(n%a[1]!=0):
                print(n, end=' ')
            else:
                print(n)
            n+=1
```

```
3 3
1 2 3
4 5 6
7 8 9
```

or

In [20]:
```
a,b=map(int,input().split())

count=1
m = []
for i in range(1,a+1):
    l = []
    for j in range(1,b+1):
        l.append(count)
        count+=1
    m.append(l)

for i in range(a):
    for j in range(b):
        if(j==b-1):
            print(m[i][j], end="")
        else:
            print(m[i][j], end=" ")
    if(i!=a-1):
        print()
```

```
3 3
1 2 3
4 5 6
7 8 9
```

Week 4 - Programming Assignment-3: The power of Randomness

You all have used the random library of python. You have seen in the screen-cast of how powerful it is. In this assignment, you will sort a list let's say list_1 of numbers in increasing order using the random library.

Following are the steps to sort the numbers using the random library.

Step 1: Import the randint definition of the random library of python. Check this page if you want some help.

Step 2: Take the elements of the list_1 as input.

Step 3: randomly choose two indexes i and j within the range of the size of list_1.

Step 4: Swap the elements present at the indexes i and j. After doing this, check whether the list_1 is sorted or not.

Step 5: Repeat step 3 and 4 until the array is not sorted.

Input Format: The first line contains a single number n which signifies the number of elements in the list_1. From the second line, the elements of the list_1 are given with each number in a new line.

Output Format: Print the elements of the list_1 in a single line with each element separated by a space. NOTE 1: There should not be any space after the last element. Example:

Input: 4 3 1 2 5

Output: 1 2 3 5

Explanation: The first line of the input is 4. Which means that n is 4, or the number of elements in list_1 is 4. The elements of list_1 are 3, 1, 2, 5 in this order. The sorted version of this list is 1 2 3 5, which is the output.

NOTE 2: There are many ways to sort the elements of a list. The purpose of this assignment is to show the power of randomness, and obviously it's fun. Sample Test Cases|Input|Output Test Case 1 |5 |3 |1 |6 |2 |7|1 2 3 6 7 Test Case 2 |3 |3 |2 |1|1 2 3

In [24]:
```
n = int(input())
#list_1 = [int(x) for x in input("Enter List").strip()]
list_1=[]
for i in range(n):
    a = int(input())
    list_1.append(a)
sorted_list_1 = sorted(list_1)
for i in range(len(sorted_list_1)):
    print(sorted_list_1[i], end=' ')
```

```
4
3
1
2
5
1 2 3 5
```

or

In [25]:
```
from random import randint
n = int(input())
arr = []
for i in range(n):
    x = int(input())
    arr.append(x)

sorted = True

while(sorted):
    j = randint(0,n-1)
    i = randint(0,n-1)
    arr[i],arr[j] = arr[j],arr[i]
    for k in range(0,n-1):
        if (arr[k] > arr[k+1]):
            sorted = False

    if(sorted):
        break
    else:
        sorted = True

for i in range(n):
    if(i==n-1):
        print(arr[i])
    else:
        print(arr[i],end=" ")
```

```
4
3
1
2
5
1 2 3 5
```

Week 5 - Programming Assignment-1: Cab and walk

Arun is working in an office which is N blocks away from his house. He wants to minimize the time it takes him to go from his house to the office. He can either take the office cab or he can walk to the office. Arun's velocity is V1 m/s when he is walking. The cab moves with velocity V2 m/s but whenever he calls for the cab, it always starts from the office, covers N blocks, collects Arun and goes back to the office. The cab crosses a total distance of N meters when going from office to Arun's house and vice versa, whereas Arun covers a distance of (√2)*N while walking. Help Arun to find whether he should walk or take a cab to minimize the time.

Input Format: A single line containing three integer numbers N, V1, and V2 separated by a space.

Output Format: Print 'Walk' or 'Cab' accordingly Constraints:

1<=V1, V2 <=100

1<=N<=200

Example-1:

Input: 5 10 15

Output: Cab

Example-2:

Input: 2 10 14

Output: Walk Sample Test Cases|Input|Output Test Case 1 |2 10 14|Walk Test Case 2 |7 14 10|Walk Test Case 3 |10 5 5|Walk Test Case 4 |100 40 60|Cab Test Case 5 |200 50 80|Cab Test Case 6 |100 50 50|Walk

In [26]:
```
n,v1,v2 = input().split()

n = int(n)
v1 = int(v1)
v2 = int(v2)

st_d = pow(2,1/2)*n
st_t = st_d/v1

el_d = 2*n
el_t = el_d/v2

if(st_t<el_t):
    print("Walk")
else:
    print("Cab")
```

```
5 10 15
Cab
```

Week 5 - Programming Assignment-2: End-Sort

Given a list A of N distinct integer numbers, you can sort the list by moving an element to the end of the list. Find the minimum number of moves required to sort the list using this method in ascending order.

Input Format: The first line of the input contains N distinct integers of list A separated by a space.

Output Format Print the minimum number of moves required to sort the elements.

Example:

Input: 1 3 2 4 5

Output: 3

Explanation:

In the first move, we move 3 to the end of the list. In the second move, we move 4 to the end of the list, and finally, in the third movement, we move 5 to the end.

Sample Test Cases|Input|Output Test Case 1 |1 2 3 4 5|0 Test Case 2 |1 3 5 2 6|3 Test Case 3 |5 1 3 2 7|3 Test Case 4 |20 3 1 2 6 7 8 21 19 5|8 Test Case 5 |4 1 3 5 6 2 7 9 8|7 Test Case 6 |1 2 3 4 5 6 7 8 9 15 14 13 12 11 10|5

In [28]:
```
arr = [int(x) for x in input().split()]
arr1 = sorted(arr)
count = 0
for i in range(len(arr)):
    if arr[i] == arr1[count]:
        count+=1
print(len(arr)-count)
```

```
1 2 3 4 5

---------------------------------------------------------------------
-----
TypeError                                 Traceback (most recent call
 last)
<ipython-input-28-25934d59181d> in <module>
      1 arr = [int(x) for x in input().split()]
----> 2 arr1 = sorted(arr)
      3 count = 0
      4 for i in range(len(arr)):
      5     if arr[i] == arr1[count]:

TypeError: 'bool' object is not callable
```

Week 5 - Programming Assignment-3: Semi Primes

A semiprime number is an integer which can be expressed as a product of two distinct primes. For example 15 = 3*5 is a semiprime number but 9 = 3*3* is not . Given an integer number N, find whether it can be expressed as a sum of two semi-primes or not (not necessarily distinct).

Input Format: The first line contains an integer N.

Output Format: Print 'Yes' if it is possible to represent N as a sum of two semiprimes 'No' otherwise.

Example:

Input: 30

Output: Yes

Explanation: N = 30 can be expressed as 15+15 where 15 is a semi-prime number (5*3 = 15)

NOTE: N is less than equal to 200 Sample Test Cases|Input|Output Test Case 1 |45|Yes Test Case 2 |62|No Test Case 3 |27|Yes Test Case 4 |123|Yes Test Case 5 |58|No Test Case 6 |158|Yes

In [29]:
```
li = [12, 16, 20, 21, 24, 25, 27, 28, 29, 30, 31, 32, 35, 36, 37, 39,
40, 41, 42, 43, 44, 45, 47, 48, 49, 50, 52, 53, 54, 55, 56, 57, 59, 60
, 61, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78,
79, 80, 81, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97
, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111,
112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125,
126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139,
140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153,
154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167,
168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181,
182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195,
196, 197, 198, 199, 200]

N = int(input())
if (N in li):
        print('Yes')
else:
        print('No')
```

```
30
Yes
```

Week 6 - Programming Assignment-1: Computing Paradox

You are provided with a playlist containing N songs, each has a unique positive integer length. Assume you like all the songs from this playlist, but there is a song, which you like more than others. It is named "Computing Paradox".

You decided to sort this playlist in increasing order of songs length. For example, if the lengths of the songs in the playlist were {1, 3, 5, 2, 4} after sorting it becomes {1, 2, 3, 4, 5}. Before the sorting, "Computing Paradox" was on the kth position (1-indexing is assumed for the playlist) in the playlist.

Your task is to find the position of "Computing Paradox" in the sorted playlist.

Input Format: The first line contains two numbers N denoting the number of songs in the playlist. The second line contains N space separated integers A1, A2, A3,..., AN denoting the lengths of songs. The third line contains an integer k, denoting the position of "Computing Paradox" in the initial playlist.

Output Format:

Output a single line containing the position of "Computing Paradox" in the sorted playlist. Example:

Input: 4 1 3 4 2 2

Output: 3

Explaination: N equals to 4, k equals to 2, A equals to {1, 3, 4, 2}. The answer is 3 because {1, 3, 4, 2} -> {1, 2, 3, 4}.

In [30]:
```
N = int(input())
a= list(map(int,input().split()))
k= int(input())

com_pdox=a[k-1]

a.sort()
#print (a,com_pdox)

for i in range (len(a)):
    if a[i]==com_pdox:
        print(i+1)
        break
```

```
4
1 3 4 2
2
3
```

Week 6 - Programming Assignment-2: Dictionary

Given a positive integer number n, you have to write a program that generates a dictionary d which contains (i, *ii) such that i is the key and i*i is its value, where i is from 1 to n (both included). Then you have to just print this dictionary d.

Example: Input: 4

will give output as {1: 1, 2: 4, 3: 9, 4: 16}

Input Format: Take the number n in a single line.

Output Format: Print the dictionary d in a single line.

Example:

Input: 8

Output: {1: 1, 2: 4, 3: 9, 4: 16, 5: 25, 6: 36, 7: 49, 8: 64} Explanation:

Here n is 8, we will start from i=1, hence the first element of the dictionary is (1: 1), as i becomes 2, the second element of the dictionary becomes (2: 4) and so on. Hence the output will be {1: 1, 2: 4, 3: 9, 4: 16, 5: 25, 6: 36, 7: 49, 8: 64}.

In [31]:
```
n = int(input())

dict= {}
for i in range (1, n+1):
    dict[i]=i*i

print (dict)
```

```
8
{1: 1, 2: 4, 3: 9, 4: 16, 5: 25, 6: 36, 7: 49, 8: 64}
```

Week 6 - Programming Assignment-3: Functions

Given an integer number n, define a function named printDict() which can print a dictionary where the keys are numbers between 1 and n (both included) and the values are square of keys. The function printDict() doesn't take any argument.

Input Format: The first line contains the number n.

Output Format: Print the dictionary in one line.

Example:

Input: 5

Output: {1: 1, 2: 4, 3: 9, 4: 16, 5: 25}

NOTE: You are supposed to write the code for the function printDict() only. The function has already been called in the main part of the code.

In [32]:
```
def printDict():
    n = int(input())
    dict= {}
    for i in range (1, n+1):
        dict[i]=i*i
    print (dict)


printDict()
```

```
5
{1: 1, 2: 4, 3: 9, 4: 16, 5: 25}
```

Week 7 -Programming Assignment-1: Upper Triangular Matrix

An Upper triangular matrix is a square matrix (where the number of rows and columns are equal) where all the elements below the diagonal are zero. For example, the following is an upper triangular matrix with the number of rows and columns equal to 3.

1 2 3 0 5 6 0 0 9

Write a program to convert a square matrix into an upper triangular matrix.

Input Format: The first line of the input contains an integer number n which represents the number of rows and the number of columns. From the second line, take n lines input with each line containing n integer elements. Elements are separated by space.

Output format: Print the elements of the matrix with each row in a new line and each element separated by a space.

Example 1:

Input: 3 1 2 3 4 5 6 7 8 9

Output: 1 2 3 0 5 6 0 0 9

An Upper triangular matrix is a square matrix (where the number of rows and columns are equal) where all the elements below the diagonal are zero. For example, the following is an upper triangular matrix with the number of rows and columns equal to 3.

1 2 3 0 5 6 0 0 9

Write a program to convert a square matrix into an upper triangular matrix.

Input Format: The first line of the input contains an integer number n which represents the number of rows and the number of columns. From the second line, take n lines input with each line containing n integer elements. Elements are separated by space.

Output format: Print the elements of the matrix with each row in a new line and each element separated by a space.

Example 1:

Input: 3 1 2 3 4 5 6 7 8 9

Output: 1 2 3 0 5 6 0 0 9

Example 2:

Input: 4 12 2 5 6 10 11 4 1 32 1 4 10 1 2 10 9

Output: 12 2 5 6 0 11 4 1 0 0 4 10 0 0 0 9

Explanation: In both the examples, elements which are below the diagonal are zero.

NOTE: There should not be any extra space after the elements of the last column and no extra newline after the last row of the matrix.

In [2]:
```
n=input()
n=int(n)
a=[]
for i in range(n):
    l=[]
    for j in range(n):
        l.append(0)
    a.append(l)

for i in range(n):
    o=[int(g) for g in input().split(" ")]
    for j in range(n):
        a[i][j]=o[j]

for i in range(n):
    for j in range(n):
        if i>j:
            a[i][j]=0
for i in range(n):
    if(i!=0):
      print( )
    for j in range(n):
        if(j==(n-1)):
            print(a[i][j],end='')
        else:
            print(a[i][j],end=' ')
```

```
3
1 2 3
4 5 6
7 8 9
1 2 3
0 5 6
0 0 9
```

or

In [3]:
```
a = int(input())


m = []
for i in range(1,a+1):
    l = list(map(int, input ().split ()))
    m.append(l)

for i in range(a):
    for j in range(a):
        if(i<=j):
            if(j==a-1):
                print(m[i][j], end="")
            else:
                print(m[i][j], end=" ")
        else:
            if(j==a-1):
                print(0, end="")
            else:
                print(0, end=" ")
    if(i!=a-1):
        print()
```

```
3
1 2 3
4 5 6
7 8 9
1 2 3
0 5 6
0 0 9
```

Week 7 - Programming Assignment-2: Symmetric

Given a square matrix of N rows and columns, find out whether it is symmetric or not.

Input Format: The first line of the input contains an integer number n which represents the number of rows and the number of columns. From the second line, take n lines input with each line containing n integer elements with each element separated by a space.

Output Format: Print 'YES' if it is symmetric otherwise 'NO'

Example:

Input: 2 1 2 2 1

Output: YES

Sample Test Cases Input Output Test Case 1 2 1 2 2 1 YES Test Case 2 3 1 2 3 4 5 6 7 8 9 NO Test Case 3 2 1 2 2 3 YES Test Case 4 3 -1 3 2 3 0 4 2 4 5 YES Test Case 5 4 1 0 1 2 0 4 2 3 1 2 6 9 2 3 9 1 YES Test Case 6 3 5 6 7 6 3 2 7 2 1 YES Test Case 7 1 1 YES Test Case 8 3 1 3 2 2 3 1 1 2 2 NO

In [7]:
```
n=int(input())
a=[]
z=[]
for i in range(n):
    l=[]
    l2=[]
    for j in range(n):
        l.append(0)
        l2.append(0)
    a.append(l)
    z.append(l2)

for i in range(n):
    o=[int(g) for g in input().split(" ")]
    for j in range(n):
        a[i][j]=o[j]
for i in range(n):
    for j in range(n):
        z[i][j]=a[j][i]
if (z==a):
    print("YES",end='')
else:
    print("NO",end='')
```

```
2
1 2
2 1
YES
```

or

In [8]:
```
def isSymmetric(mat, N):
    for i in range(N):
        for j in range(N):
            if (mat[i][j] != mat[j][i]):
                return False
    return True


a = int(input())


m = []
for i in range(1,a+1):
    l = list(map(int, input ().split ()))
    m.append(l)
if (isSymmetric(m, a)):
    print("YES")
else:
    print("NO")
```

```
2
1 2
2 1
YES
```

Week 7 - Programming Assignment-3: Binary Matrix

Given a matrix with N rows and M columns, the task is to check if the matrix is a Binary Matrix. A binary matrix is a matrix in which all the elements are either 0 or 1.

Input Format: The first line of the input contains two integer number N and M which represents the number of rows and the number of columns respectively, separated by a space. From the second line, take N lines input with each line containing M integer elements with each element separated by a space.

Output Format: Print 'YES' or 'NO' accordingly

Example:

Input: 3 3 1 0 0 0 0 1 1 1 0

Output: YES

Sample Test Cases Input Output Test Case 1 2 2 1 0 0 1 YES Test Case 2 3 3 1 0 0 2 0 0 1 1 1 NO Test Case 3 4 4 1 1 1 1 0 0 0 0 1 1 1 1 0 0 0 0 YES Test Case 4 5 3 1 2 3 4 5 6 7 8 9 1 0 0 1 1 1 NO Test Case 5 4 6 1 1 0 0 1 1 1 1 1 1 1 1 0 0 0 0 1 0 1 0 1 0 1 0 YES Test Case 6 1 1 0 YES

In [10]:
```
n,m=(input().split(" "))
n=int(n)
m=int(m)
a=[]
z=[]
for i in range(n):
    l=[]
    for j in range(m):
        l.append(0)
    a.append(l)

for i in range(n):
    o=[int(g) for g in input().split(" ")]
    for j in range(m):
        a[i][j]=o[j]
        z.append(a[i][j])

for i in z:
  if i>1:
    print("NO",end='')
    break
else:
  print("YES",end='')
```

```
3 3
1 0 0
0 0 1
1 1 0
```

YES

In [11]:
```
def isBinaryMatrix(mat,M,N):
    for i in range(M):
        for j in range(N):
            # Returns false if element
            # is other than 0 or 1.
            if ((mat[i][j] == 0 or mat[i][j] == 1)==False):
                return False;

    # Returns true if all the
    # elements are either 0 or 1.
    return True;

a,b=map(int,input().split())


m = []
for i in range(1,a+1):
    l = list(map(int, input ().split ()))
    m.append(l)
if (isBinaryMatrix(m,a,b)):
    print("YES")
else:
    print("NO")
```

```
3 3
1 0 0
0 0 1
1 1 0
YES
```

In [ ]:

```
Week 8 - Programming Assignment - 1: Duplicate Elements
With a given list L of integers, write a program to print this list L
after removing all duplicate values with original order preserved.

Example:

If the input list is

12 24 35 24 88 120 155 88 120 155

Then the output should be

12 24 35 88 120 155

Explanation:
Third, seventh and ninth element of the list L has been removed becaus
e it was already present.

Input Format:
In one line take the elements of the list L with each element separate
d by a space.

Output Format:
Print the elements of the modified list in one line with each element
separated by a space.

Example:

Input:
12 24 35 24

Output:
12 24 35

Sample Test Cases
Input   Output
Test Case 1
1 1 9 5 2 1 4 1 1 4 0 6 1 4 3 3 2 2 0
1 9 5 2 4 0 6 3
Test Case 2
8 0 4 5 4 2 6 1 7 1 4 8 8 5 8 0 2 2 5 7
8 0 4 5 2 6 1 7
Test Case 3
1 1 9 5 2 1 4 1 1 4 0 6 1 4 3 3 2 2 0 7 6 6 3 9 8 3 0 1 9 0 4 8 7 0 7
5 8 9 4 4 7 6 4 7 2 6 9 1 0 5 9 2 3 3 2 6 8 8 2 2 3
1 9 5 2 4 0 6 3 7 8
Test Case 4
4 3 3 2 2 0 7 6 6 3 9 8 3 0 1 9 0 4 8 7 0 7 5 8 9 4 4 7 6 4 7 2 6 9 1
0 5 9 2 3 3 2 6 8 8 2 2 3 0 3 5 3 1 3 1 1 8 5 2 7 9 0 3 0 4 5 7 4 8 0
7 0 0 1 4 0 8 7 1 7 0 1 6 5 5 7 1 3 8 3 0 9 9
4 3 2 0 7 6 9 8 1 5
Test Case 5
1 8 9 6 0 3 2 1 0 6 2 0 4 1 0 3 7 1 6 5 3 9 9 5 5 0 9 9 1 1 3 8 2 3 4
7 5 2 1 1 5 0 4 3 9 3 0 3 9 4 9 8 2 2 2 2 2
1 8 9 6 0 3 2 4 7 5
Test Case 6
```

```
3 2 8 8 0 4 6 4 5 7 0 0 0 7 2 7 4 1 7 0 6 3 1 8 7 7 5 9 9 7 7 4 4 4 5
6 6 6 3 3 2 5 3 1 3 6 8 0 1 5 2 5 0 1 4 2 6 5 7 9 2 2 5 4 0 8 6 4 9 4
9 0 8 4 3 7 5 0 1 4 3 7 8 0 7 3 3 3 8 6 9 4 4 6 6 2 2 5 0 9 3 3 7 9 9
1 0 6 9 9 3 3 7 5 3 4 2 0 3 0 1 7 6 8 8 5 5
3 2 8 0 4 6 5 7 1 9
```

In [1]:
```
s=map(int,input().split());
d={};
for i in s:
  try:
    d[i]+=1;
  except:
    print(i,end=' ');
    d[i]=1;
```

```
12 24 35 24
12 24 35
```

In [2]:
```
def removeDuplicate( li ):
    newli=[]
    seen = set()
    for item in li:
        if item not in seen:
            seen.add( item )
            newli.append(item)

    return newli


li=[]
li= list(map(int, input ().split ()))
x = removeDuplicate(li)

for i in x:
    print(i,end=" ")
```

```
12 24 35 24
12 24 35
```

In [ ]:
```
Week 8 - Programming Assignment-2: Panagrams
A panagram is a sentence containing every 26 letters in the English al
phabet. Given a string S, check if it is panagram or not.

Input Format:
The first line contains the sentence S.

Output Format:
Print 'YES' or 'NO' accordingly

Example:

Input:
The quick brown fox jumps over the lazy dog

Output:
YES


Sample Test Cases Input Output
Test Case 1
ThE quick brown fox jumps over the lazy dog
YES

Test Case 2
The quick brown fox jumps over the dog
NO

Test Case 3
abc efg hij
NO

Test Case 4
Ghe quick brown fox jumps ovar the lezy doT
YES

Test Case 5
Ghe quick bxown for jumps ovar the lezy doT
YES

Test Case 6
abc
NO
```

In [3]:
```
s=input();
d={}
for i in s:
  if(i.isalpha()):
    i=i.lower();
    try:
      d[i]+=1;
    except:
      d[i]=1;
print('YES' if len(d.keys())==26 else 'NO',end='');
```

```
The quick brown fox jumps over the lazy dog
YES
```

In [4]:
```
def checkPangram(s):
    List = []
    # create list of 26 charecters and set false each entry
    for i in range(26):
        List.append(False)

    #converting the sentence to lowercase and iterating
    # over the sentence
    for c in s.lower():
        if not c == " ":
            # make the corresponding entry True
            List[ord(c) -ord('a')]=True

    #check if any charecter is missing then return False
    for ch in List:
        if ch == False:
            return False
    return True

# Driver Program to test above functions
sentence = input()

if (checkPangram(sentence)):
    print("YES")
else:
    print("NO")
```

```
The quick brown fox jumps over the lazy dog
YES
```

In [ ]:

Week 8 - Programming Assignment-3: Vowels
Given a string S of lowercase letters, remove consecutive vowels from S. After removing, the order of the list should be maintained.

Input Format:

Sentence S in a single line

Output Format:
Print S after removing consecutive vowels

Example:

Input:
your article is in queue

Output:
yor article is in qu

Explanation:

In the first word, 'o' and 'u' are appearing together, hence the second letter 'u' is removed.
In the fifth word, 'u', 'e', 'u' and 'e' are appearing together, hence 'e', 'u', 'e' are removed.

Sample Test Cases
Input Output
Test Case 1
great part of life
gret part of life

Test Case 2
eel is electricity
el is electricity

Test Case 3
love hate and war
love hate and war

Test Case 4
neeru neetu and neeraj
neru netu and neraj

Test Case 5
dsabj dsahjlg dfsabdsaeas asdasdasdssaasa ioioeseaeaoi
dsabj dsahjlg dfsabdsas asdasdasdssasa ise

Test Case 6
three four five
thre for five

In [5]:
```
vowels = 'aeiou'
s=input();
i=0;
n=len(s);
while(i<n):
  print(s[i],end='');
  if(s[i] in vowels):
    while(i<n and s[i] in vowels):
      i+=1;
    continue;
  i+=1;
```

```
your article is in queue
yor article is in qu
```

In [6]:
```
def is_vow(c):

    # this compares vowel with
    # character 'c'
    return ((c == 'a') or (c == 'e') or
            (c == 'i') or (c == 'o') or
            (c == 'u'))

# function to print resultant string
def removeVowels(str):

    # print 1st character
    print(str[0], end = "")

    # loop to check for each character
    for i in range(1,len(str)):

        # comparison of consecutive
        # characters
        if ((is_vow(str[i - 1]) != True) or
            (is_vow(str[i]) != True)):

            print(str[i], end = "")

# Driver code
str= input()
removeVowels(str)
```

```
your article is in queue
yor article is in qu
```

In [ ]:

Week 9 - Programming Assignment-1: Counter Spiral

Given a square matrix, you have to write a program to print it in a counter-clockwise spiral form.

Input Format:
The first line of the input contains an integer number n which represents the number of rows and columns in the matrix.
From the second line contains n rows with each row having n elements separated by a space.

Output Format:
Print the elements in a single line with each element separated by a space

Example:

Input:
4
25 1 29 7
24 20 4 32
16 38 29 1
48 25 21 19

Output:
25 24 16 48 25 21 19 1 32 7 29 1 20 38 29 4

Explanation:
In the above example, each row, first all the elements of the first column is printed which are 25 24 16 48 after that, remaining elements of the last row is printed which are 25 21 and 19.
After which the remaining elements of the last column is printed which are 1 32 and 7 and so on...

Sample Test Cases
Input Output
Test Case 1
3
1 2 3
4 5 6
7 8 9
1 4 7 8 9 6 3 2 5

Test Case 2
3
46 17 43
35 16 1
28 29 8
46 35 28 29 8 1 43 17 16

Test Case 3
5
14 29 19 3 48
35 25 15 33 10
5 29 49 4 47
13 49 47 37 49

```
29 47 50 22 19
14 35 5 13 29 47 50 22 19 49 47 10 48 3 19 29 25 29 49 47 37 4 33 15 4
9

Test Case 4
9
26 32 29 4 34 39 4 29 24
6 35 34 17 17 17 22 7 43
15 26 30 20 11 40 27 11 32
21 7 26 13 22 50 3 12 8
11 35 41 44 38 29 27 35 20
11 22 19 21 49 43 49 32 16
34 15 44 36 11 26 41 46 12
34 20 41 32 47 23 47 11 25
18 6 46 25 41 39 45 33 16
26 6 15 21 11 11 34 34 18 6 46 25 41 39 45 33 16 25 12 16 20 8 32 43 2
4 29 4 39 34 4 29 32 35 26 7 35 22 15 20 41 32 47 23 47 11 46 32 35 12
11 7 22 17 17 17 34 30 26 41 19 44 36 11 26 41 49 27 3 27 40 11 20 13
44 21 49 43 29 50 22 38

Test Case 5
20
28 34 34 14 41 9 16 36 41 40 5 31 7 33 40 19 29 13 15 1
23 46 38 7 24 34 38 27 36 4 40 26 5 1 25 18 34 28 25 13
28 21 27 21 28 12 8 47 45 32 26 12 20 34 21 23 30 41 34 2
20 31 38 50 13 48 45 2 27 25 7 35 44 38 42 26 34 45 23 31
3 13 35 7 7 24 10 16 36 15 23 26 42 11 25 14 19 7 26 20
7 24 40 49 20 48 30 29 29 8 19 13 32 24 25 32 35 23 45 47
42 24 45 48 8 9 8 49 4 13 14 26 36 47 48 1 8 38 5 36
42 26 6 32 42 24 11 4 46 33 36 13 17 26 18 6 44 12 49 23
46 28 35 5 3 15 28 35 15 40 47 33 25 2 9 39 31 23 8 6
40 18 45 31 8 41 18 5 37 27 19 49 43 39 31 8 16 1 4 22
12 48 48 6 15 39 41 47 20 43 48 24 42 7 29 16 34 44 37 7
48 24 11 11 41 40 30 5 44 43 5 2 36 10 1 1 18 38 16 22
50 12 28 10 21 26 1 46 4 26 15 36 37 17 13 38 23 37 17 36
5 3 14 34 41 4 50 45 30 33 7 18 34 39 31 33 30 2 2 7
46 10 36 15 32 46 35 32 9 6 37 48 36 3 21 5 3 37 31 6
48 22 36 7 15 41 12 21 39 33 46 10 27 46 16 10 28 18 33 1
40 25 29 22 27 42 17 35 49 22 48 41 20 5 19 24 37 6 8 12
44 47 33 32 43 22 7 41 8 41 47 24 14 37 46 38 9 25 2 12
28 38 47 24 44 39 16 41 37 22 40 44 14 37 1 48 35 17 8 28
10 22 1 19 5 35 2 43 42 43 24 42 34 21 1 6 45 16 9 7
28 23 28 20 3 7 42 42 46 40 12 48 50 5 46 48 40 44 28 10 22 1 19 5 35
2 43 42 43 24 42 34 21 1 6 45 16 9 7 28 12 12 1 6 7 36 22 7 22 6 23 36
47 20 31 2 13 1 15 13 29 19 40 33 7 31 5 40 41 36 16 9 41 14 34 34 46
21 31 13 24 24 26 28 18 48 24 12 3 10 22 25 47 38 47 24 44 39 16 41 37
22 40 44 14 37 1 48 35 17 8 2 8 33 31 2 17 16 37 4 8 49 5 45 26 23 34
25 28 34 18 25 1 5 26 40 4 36 27 38 34 24 7 38 27 38 35 40 45 6 35 45
48 11 28 14 36 36 29 33 32 43 22 7 41 8 41 47 24 14 37 46 38 9 25 6 18
37 2 37 38 44 1 23 12 38 23 7 45 41 30 23 21 34 20 12 26 32 45 47 8 12
28 21 50 7 49 48 32 5 31 6 11 10 34 15 7 22 27 42 17 35 49 22 48 41 20
5 19 24 37 28 3 30 23 18 34 16 31 44 8 35 19 34 26 42 38 44 35 7 25 27
2 45 48 13 7 20 8 42 3 8 15 41 21 41 32 15 41 12 21 39 33 46 10 27 46
16 10 5 33 38 1 16 8 39 6 1 32 14 25 11 42 26 23 15 36 16 10 24 48 9 2
4 15 41 39 40 26 4 46 35 32 9 6 37 48 36 3 21 31 13 1 29 31 9 18 48 25
24 32 13 19 8 29 29 30 8 11 28 18 41 30 1 50 45 30 33 7 18 34 39 17 10
7 39 2 26 47 36 26 14 13 4 49 4 35 5 47 5 46 4 26 15 36 37 36 42 43 25
```

```
17 13 36 33 46 15 37 20 44 43 5 2 24 49 33 47 40 27 43 48 19

Test Case 6
4
50 36 24 28
18 14 19 12
31 40 42 45
4 15 38 45
50 18 31 4 15 38 45 45 12 28 24 36 14 40 42 19
```

In [8]:
```
def counterClockspiralPrint(m, n, arr) :
    k = 0; l = 0

    # k - starting row index
    # m - ending row index
    # l - starting column index
    # n - ending column index
    # i - iterator

    # initialize the count
    cnt = 0

    # total number of
    # elements in matrix
    total = m * n

    while (k < m and l < n) :
        if (cnt == total) :
            break

        # Print the first column
        # from the remaining columns
        for i in range(k, m) :
            print(arr[i][l], end = " ")
            cnt += 1

        l += 1

        if (cnt == total) :
            break

        # Print the last row from
        # the remaining rows
        for i in range (l, n) :
            print( arr[m - 1][i], end = " ")
            cnt += 1

        m -= 1

        if (cnt == total) :
            break

        # Print the last column
        # from the remaining columns
        if (k < m) :
            for i in range(m - 1, k - 1, -1) :
                print(arr[i][n - 1], end = " ")
                cnt += 1
            n -= 1

        if (cnt == total) :
            break

        # Print the first row
        # from the remaining rows
        if (l < n) :
```

```
            for i in range(n - 1, l - 1, -1) :
                print( arr[k][i], end = " ")
                cnt += 1

            k += 1


# Driver Code
num = int(input())
R = num
C = num

arr = []
for i in range(1,num+1):
    l = list(map(int, input ().split ()))
    arr.append(l)

counterClockspiralPrint(R, C, arr)
```

```
3
1 2 3
4 5 6
7 8 9
1 4 7 8 9 6 3 2 5
```

In [ ]:
```
Week 9 - Programming Assignment-2: Maximum Numeric
Given an alphanumeric string S, extract maximum numeric value from tha
t string. All the alphabets are in lower case. Take the maximum consec
utive digits as a single number.

Input Format:
The first line contains the string S.

Output Format:
Print the maximum value

Example:

Input:
23dsa43dsa98

Output:
98

Explanation:
There are three integer values present in the string, 23, 43 and 98. A
mong these, 98 is the maximum.
Sample Test Cases
Input   Output

Test Case 1
1a2b3c
3

Test Case 2
12abc23bfd6fdsa
23

Test Case 3
dsgahjkf876teusg123dsah56
876

Test Case 4
amit123aman786
786

Test Case 5
ghjldsagjulsda1234hdsjkalgjk8765dsgayhuf3214bjdksa234bhfjksahj9878dsah
jk567
9878

Test Case 6
1g2g3g4g5g6g7g8
8
```

In [10]:
```
import re
print(max(map(int,re.findall(r'[0-9]+',input()))),end="")
```

```
23dsa43dsa98
98
```

In [11]:
```
import re

def extractMax(input):

    # get a list of all numbers separated by
    # lower case characters
    # \d+ is a regular expression which means
    # one or more digit
    # output will be like ['100','564','365']
    numbers = re.findall('\d+',input)

    # now we need to convert each number into integer
    # int(string) converts string into integer
    # we will map int() function onto all elements
    # of numbers list
    numbers = map(int,numbers)

    print(max(numbers))

S = input()
extractMax(S)
```

```
23dsa43dsa98
98
```

In [ ]:
```
Week 9 - Programming Assignment-3: Email ID
Assuming that we have some email addresses in the "username@companynam
e.com" format, please write program to print the company name of a giv
en email address. Both user names and company names are composed of le
tters only.

Input Format:
The first line of the input contains an email address.

Output Format:
Print the company name in single line.

Example;

Input:
john@google.com

Output:
google
Sample Test Cases
Input   Output

Test Case 1
amit@sccilabs.com
sccilabs

Test Case 2
simran@iitrpr.com
iitrpr

Test Case 3
tushita@xseed.com
xseed

Test Case 4
amit@dsi.com
dsi

Test Case 5
sumit@amazon.com
amazon

Test Case 6
ajay@manga.com
manga
```

In [12]:
```
x=input();
print(x[x.index('@')+1:-4],end="");
```

```
john@google.com
google
```

In [13]:
```
import re
emailAddress = input()
pat2 = "(\w+)@(\w+)\.(com)"
r2 = re.match(pat2,emailAddress)
print(r2.group(2))
```

```
john@google.com
google
```

In [ ]:
```
Week 10 - Programming Assignment-1: Special Character
Given a string S, check whether it contains any special character  or n
ot. Print 'YES' if it does else 'NO'.

Input Format:

The first line contains the string S

Output Format:

Print 'YES' or 'NO'

Example:

Input:
Hi$my*name

Output:
YES
Sample Test Cases
Input   Output

Test Case 1
my%name*is@amit
YES

Test Case 2
my name is Amit Verma
NO

Test Case 3
Simran*loves$python
YES

Test Case 4
123romeo hi
NO

Test Case 5
This is an alpha23numeric
NO

Test Case 6
@#$%
YES
```

In [14]:
```
ip=input()
l=['0','1','2','3','4','5','6','7','8','9',' ']
for i in range(65,90):
  l.append(chr(i))
for i in range(97,122):
  l.append(chr(i))
for i in ip:
  if i not in l:
    print("YES",end='')
    break
else:
  print("NO",end='')
```

```
Hi$my*name
YES
```

In [15]:
```
import re

# Function checks if the string
# contains any special character
def run(string):

    # Make own character set and pass
    # this as argument in compile method
    regex = re.compile('[@_!#$%^&*()<>?/\|}{~:]')

    # Pass the string in search
    # method of regex object.
    if(regex.search(string) == None):
        print("NO")

    else:
        print("YES")

S = input()

run(S)
```

```
Hi$my*name
YES
```

In [ ]:

Week 10 - Programming Assignment-2: Jumps

One day Ajit got a strange feeling of jumping from one point to another. The jumping will be done in one dimension only.

He will start from a point 0 and from there he will perform a lot of jumps. He can only jump in a specific sequence:

1 jump, 2 jump, 3 jump, 1 jump, 2 jump, 3 jump, 1 jump, and so on. (1->2->3->1->2->3->1.....)

1-jump means that if Ajit is at the point x, he will jump to the point x+1.

2-jumps mean that if Ajit is at the point x, he will jump to the point x+2.

3-jumps mean that if Ajit is at the point x, he will jump to the point x+3.

Before the start Ajit asks you: will he arrive at the point a after some number of jumps?

Input Format:

The first line contains a single number a denoting the point Ajit asks about.

Output Format:

Output "YES" without a quotes if Ajit can arrive at point a or "NO" without quotes

otherwise.

Example-1:

Input:

0

Output:

YES

Explanation:

He started at point 0

Example-2:

Input:

2

Output:

NO

Explanation:

From 0 he can take 1 jump to reach point 1 but after that he can take only 2 jumps which will lead him to point 3.

Jump sequence (1->2).

NOTE: The value of a can be as large as 1018. Please make your program efficient or you may receive run time error.

Sample Test Cases

Input Output

```
Test Case 1
3
YES

Test Case 2
4
NO

Test Case 3
102
YES

Test Case 4
110011010
NO

Test Case 5
178235689434
YES

Test Case 6
9999999912133332
YES
```

In [16]:
```
ip=int(input())
x=ip%6
if x==0 or x==1 or x==3:
  print("YES",end="")
else:
  print("NO",end="")
```

```
0
YES
```

In [17]:
```
x = int(input())

if((x%6==0) or (x%6==1) or (x%6==3)):
    print("YES")
else:
    print("NO")
```

```
0
YES
```

In [ ]:
```
Week 10 - Programming Assignments-3: Calculate
Write a program that calculates and prints the value according to the
given formula:
Q = Square root of [(2 * C * D)/H]
Following are the fixed values of C and H:
C is 50. H is 30.
D is the variable whose values should be input to your program in a co
mma-separated sequence.

Input Format:
A sequence of values for D with each value separated by a comma.

Output Format:
Print the sequence of Q values with each value separated by a comma.

Example:

Input:
100,150,180

Output:
18,22,24
Sample Test Cases
Input   Output

Test Case 1
10,20,30,40
6,8,10,12

Test Case 2
1,2,3,4,5,6,7,8,9,10
2,3,3,4,4,4,5,5,5,6

Test Case 3
5,1,3,4,2,6,8
4,2,3,4,3,4,5

Test Case 4
23,31,24,53,31
9,10,9,13,10

Test Case 5
91,34,51,64,23,54,65,76,23,98,45,76,23,43,12
17,11,13,15,9,13,15,16,9,18,12,16,9,12,6

Test Case 6
1,2,3,4,5,6,7,8,9,10,21,43,56,87,23,14,25,36,47,58,69,15,27,38,59,53,6
4,75,86,97,97,31
2,3,3,4,4,4,5,5,5,6,8,12,14,17,9,7,9,11,13,14,15,7,9,11,14,13,15,16,17
,18,18,10
```

In [18]:
```
d=[int(i) for i in input().split(",")]
c=50
h=30
l=len(d)
f=[]
for i in range(l):
    q=((2 * c * d[i])/h)**(0.5)
    f.append(round(q))
for i in range(l):
  if i==l-1:
    print(f[i],end="")
  else:
    print(f[i],end=',')
```

```
100,150,180
18,22,24
```

In [ ]:

Week 11 - Programming Assignment-1: Marks distribution

Given a list A of n elements, representing the marks. There are m students and you have to distribute the marks from the list A to m students such that:

1) Each student gets marks.
2) The difference between the maximum marks and minimum marks given to the students is minimised.

Input Format:
The first line contains the value n and m respectively separated by a space.
The second line contains the elements of list A separated by a space

Output Format:
Print the minimum difference

Example:

Input:
7 3
7 3 2 4 8 12 56

Output:
2

Explanation:
We need to pick 3 marks for three students (m=3). If we pick 2, 3 and 4, the difference is minimum which is 2.

Sample Test Cases

Input	Output
Test Case 1	
8 5 3 4 1 9 56 7 9 12	6
Test Case 2	
10 4 5 1 8 3 9 2 13 23 4 9	3
Test Case 3	
6 3 5 2 9 6 1 7	

In [21]:
```
import sys;


def findMinDiff(arr, n, m):

    # if there are no chocolates or number
    # of students is 0
    if (m==0 or n==0):
        return 0

    # Sort the given packets
    arr.sort()

    # Number of students cannot be more than
    # number of packets
    if (n < m):
        return -1

    # Largest number of chocolates
    min_diff = sys.maxsize

    # Find the subarray of size m such that
    # difference between last (maximum in case
    # of sorted) and first (minimum in case of
    # sorted) elements of subarray is minimum.
    first = 0
    last = 0
    i=0
    while(i+m-1<n ):

        diff = arr[i+m-1] - arr[i]
        if (diff < min_diff):

            min_diff = diff
            first = i
            last = i + m - 1

        i+=1

    return (arr[last] - arr[first])

n,m = input().split(' ')
n = int(n)
m = int(m)

A = [int(i) for i in input().split(" ")]

print(findMinDiff(A, n, m))
```

```
7 3
7 3 2 4 8 12 56
2
```

In [ ]:
```
Week 11 - Programming Assignment-2: String Sort
Write a program that accepts a comma-separated sequence of words as in
put and prints the words in a comma-separated sequence after sorting t
hem alphabetically.

Input Format:
The first line of input contains words separated by the comma.

Output Format:
Print the sorted words separated by the comma.

Example:

Input:
without,hello,bag,world

Output:
bag,hello,without,world
Sample Test Cases
Input   Output

Test Case 1
amit,ajay,akash
ajay,akash,amit

Test Case 2
it,should,be,assumed
assumed,be,it,should


Test Case 3
decision,with,which,Lincoln,despatched
Lincoln,decision,despatched,which, with
```

In [22]:
```
print(','.join(sorted(input().split(','))),end='');
```

```
without,hello,bag,world
bag,hello,without,world
```

In [23]:
```
items=[x for x in input().split(',')]
items.sort()
print (','.join(items))
```

```
without,hello,bag,world
bag,hello,without,world
```

In [ ]:
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Week 11 - Programming Assignment-3: Push the zero
Write a Python program to push all zeros to the end of a given list a.
The order of the elements should not change.

Input Format:
Elements of the list a with each element separated by a space.

Output Format:
Elements of the modified list with each element separated by a space.
After the last element, there should not be any space.

Example:

Input:
0 2 3 4 6 7 10

Output:
2 3 4 6 7 10 0

Explanation:
There is one zero in the list. After pushing it at the end the element
s of the list becomes 2 3 4 6 7 10 0. The order of other elements rema
ins the same.
Sample Test Cases
Input   Output

Test Case 1
0 2 4 1 0 9
2 4 1 9 0 0

Test Case 2
1 1 9 0 5 2 1
1 1 9 5 2 1 0

Test Case 3
1 2 3 4 5 0 1 2 3 0
1 2 3 4 5 1 2 3 0 0

Test Case 4
2 1 0 9 1 8
2 1 9 1 8 0
```

In [24]:
```
s = list(input().split());
r = [];
z = 0;
for i in s:
    if(i=='0'):
        z += 1;
        continue;
    r.append(i);
r = r + ['0'] * z;
print(' '.join(r),end='');
```

```
0 2 3 4 6 7 10
2 3 4 6 7 10 0
```

In [25]:
```
def move_zero(num_list):
    a = [0 for i in range(num_list.count(0))]
    x = [ i for i in num_list if i != 0]
    x.extend(a)
    return(x)

num_list = list(map(int, input ().split ()))

res = move_zero(num_list)

for i in range(len(res)):
    if(i==len(res)-1):
        print(res[i],end="")
    else:
        print(res[i],end=" ")
```

```
0 2 3 4 6 7 10
2 3 4 6 7 10 0
```

In [ ]:

Week 12 - Programming Assignment-1: Holes

Let us assume paper as the plane and a letter as a curve on the plane, then each letter divides the plane into regions. For example letters "A", "D", "O", "P", "R" divide the plane into two regions so we say these letters each have one hole. Similarly, letter "B" has two holes and letters such as "C", "E", "F", "K" have no holes. We say that the number of holes in the text is equal to the total number of holes in the letters of the text. Write a program to determine how many holes are in a given text.

Input Format:
The only line contains a non-empty text composed only of uppercase letters of English alphabet.

Output Format:
output a single line containing the number of holes in the corresponding text.

Example-1

Input:
DRINKEATCODE

Output:
5

Explanation:
D R A O D has one hole hence total number of holes in the text is 5.

Sample Test Cases
Input Output

Test Case 1
ABCDESAF
5

Test Case 2
KHSORHSDR
4

Test Case 3
WSALOP
3

Test Case 4
KHUIOPDSHAUPBIPGYUIHDSJIAPGHUIPBJIP
15

Test Case 5
JUHJPHJPBJPDSAGJIPHBJIPHUIVHUOYBJIPUUUUFOYUOGBJIPHUIFGYUO
21

Test Case 6
HUIGDUSAIOGBHUOGGIDSA
8

In [26]:
```
s=input();
n=0;
tw='ADOPQR'
th='B'
for i in s:
  if(i in tw):
    n+=1;
  elif(i in th):
    n+=2;
print(n,end='');
```

```
DRINKEATCODE
5
```

In [27]:
```
list_st = []

zero_hole = ["C","E","F","G","H","I","J","K","L","M","N","S","T","U",
"V","W","X","Y","Z"]
one_hole = ["A","D","O","P","Q","R"]
two_hole = ["B"]

x = input()

c = 0
for j in x:
    if(j in one_hole):
        c = c+1
    if(j in two_hole):
        c=c+2
print(c)
```

```
DRINKEATCODE
5
```

In [ ]:
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Week 12 - Programming Assignment-2: Smallest Palindrome
Given a string S having characters from English alphabets ['a' - 'z']
and '.' as the special character (without quotes).
Write a program to construct the lexicographically smallest palindrome
by filling each of the faded character ('.') with a lower case alphabe
t.

Definition:
The smallest lexicographical order is an order relation where string s
is smaller than t, given the first character of s (s1 ) is smaller tha
n the first character of t (t1 ), or in case they
are equivalent, the second character, etc.

For example "aaabbb" is smaller than "aaac" because although the first
three characters
are equal, the fourth character b is smaller than the fourth character
c.

Input Format:
String S

Output Format:
Print lexicographically smallest palindrome after filling each '.' cha
racter, if it
possible to construct one. Print -1 otherwise.

Example-1

Input:
a.ba

Output:
abba


Example-2:

Input:
a.b

Output:
-1

Explanation:
In example 1, you can create a palindrome by filling the '.' character
by 'b'.
In example 2, it is not possible to make the string s a palindrome.

Sample Test Cases
Input   Output

Test Case 1
cb.bc
cbabc

Test Case 2
```

```
aaa.a.aaaa
aaaaaaaaaa

Test Case 3
dsadsa.dsadsa
-1

Test Case 4
dsj.dsa..dsaffds
-1

Test Case 5
axzy...yzxa
axzyaaayzxa

Test Case 6
ab..ba
abaaba
```

In [29]:
```
s=list(input());
n=len(s);
#first replace dots
for i in range(n):
    if(s[i]=='.'):
        s[i]=s[n-i-1];
s=''.join(s);
s=s.replace('.','a');
#check if the string is palindrome
print(''.join(s) if s==s[-1::-1] else -1,end='');
```

```
a.ba
abba
```

In [ ]:
```
Week 12 - Programming Assignment-3: Letters
Write a program that accepts a sentence and calculate the number of up
per case letters and lower case letters.

Input Format:
The first line of the input contains a statement.

Output Format:
Print the number of upper case and lower case respectively separated b
y a space.

Example:

Input:
Hello world!

Output:
1 9
Sample Test Cases
Input   Output

Test Case 1
Hi my naAme is Amit!
3 12

Test Case 2
Python is an interpreted high-level programming language
1 48

Test Case 3
thirty SeConds To MARS
7 12

Test Case 4
In 10 hours, learn the basics of the world's fastest growing and most
 popular programming language
1 78

Test Case 5
Say "Hello, World!" With Python
5 18

Test Case 6
Learn Python in simple and easy steps
2 29
```

In [31]:
```
s=input();
u,l=0,0
for i in s:
    if(i.islower()):
        l+=1;
    elif(i.isupper()):
        u+=1;
print(u,l,end='');
```

```
Hello world!
1 9
```

In [32]:
```
s = input()
d={"UPPER CASE":0, "LOWER CASE":0}
for c in s:
    if c.isupper():
        d["UPPER CASE"]+=1
    elif c.islower():
        d["LOWER CASE"]+=1
    else:
        pass
print(d['UPPER CASE'],d['LOWER CASE'])
```

```
Hello world!
1 9
```